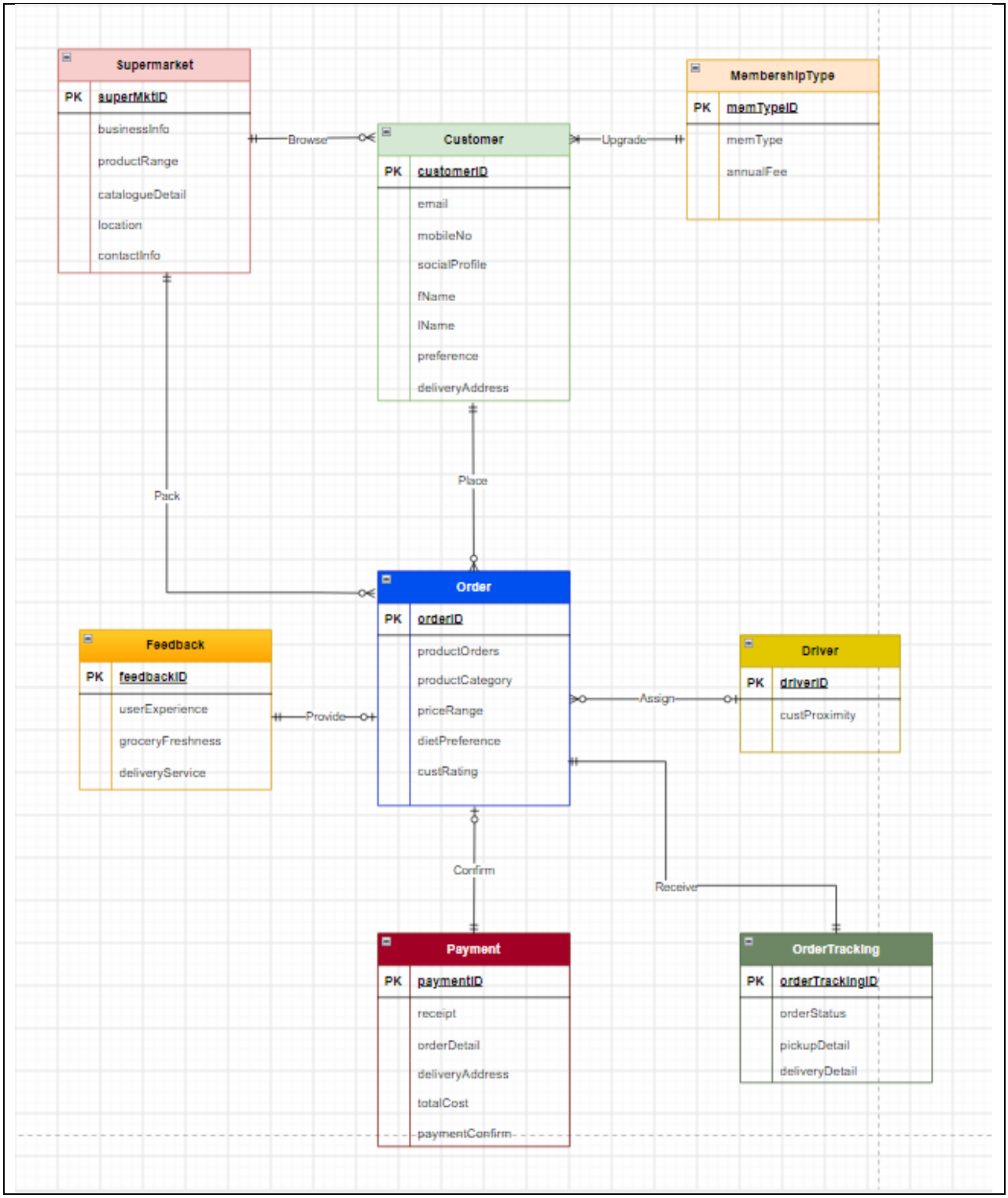
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Assessment 2: Case Study

***Task 1**: Create A Conceptual Data Model (**ERD**)*

CONCEPTUAL RELATIONAL DATA DIAGRAM



Entities	Supermarket, Customer, MembershipType, Order, Feedback, Driver, Payment, OrderTracking.

Attributes	• **Supermarket:** businessInfo, productRange, catalogueDetail, location, contactInfo • **Customer:** email, mobileNo, socialProfile, fName, lName, preference, deliveryAddress • **MembershipType:** memType, annualFee • **Order:** productOrders, productCategory, priceRange, dietPreference, custRating • **Driver:** custProximity • **Feedback:** userExperience, groceryFreshness, deliveryService • **Payment:** receipt, orderDetail, deliveryAddress, totalCost, paymentConfirm • **OrderTracking**: orderStatus, pickupDetail, deliveryDetail.
Primary Keys	supermktID (**Supermarket**), customerID (**Customer**), memTypeID (**MembershipType**), orderID (**Order**), feedbackID (**Feedback**), driverID (**Driver**), paymentID (**Payment**), orderTrackingID (**OrderTracking**).
Relationships	Browse, Pack, Receive, Upgrade, Place, Provide, Assign, Confirm, Receive.
Multiplicity (Cardinality)	(0,M) **Customer** browse **Supermarket** (1,1) (1,M) **Customer** upgrades **MembershipType** (1,1) (1,1) **Customer** places **Order** (0,M) (0,1) **Order** provide **Feedback** (1,1) (0,M) **Order** assigns **Driver** (0,1) (1,1) **Order** receive **OrderTracking** (1,1) (0,1) **Order** confirm **Payment** (1,1)

Assumptions	When working on the diagram I have decided to follow the logic of how the relationship is laid out in the diagram as I decided to follow the text in the scenario and explain what is occurring in the diagram such as a supermarket being browsed by 0 or many customers. There were some descriptions in the text such as the scenario describing supermarket, department, grocery products in an order, or how member ship have different types, but I decided to just shorten it to 'productOrders' and 'memType' to make it clearer and cleaner in the entities.

***Task 2**: Create a logical/relational data model*

CONCEPTUAL RELATIONAL DATA DIAGRAM

Supermarket (superMktID, businessInfo, productRange, catalogueDetail, location, contactInfo) **Primary Key**: superMktID **Foreign Key**: None
Customer (customerID, email, mobileNo, socialProfile, details, fName, lName, preference, deliveryAddress) **Primary Key**: customerID

Foreign Key: superMktID, references **Supermarket** (superMktID) **Foreign Key**: memTypeID, references **MembershipType** (memTypeID)
MembershipType (memTypeID, memType, annualFee) **Primary Key**: memTypeID **Foreign Key**: None
Order (orderID, productOrders, productCategory, priceRange, dietPreference, custRating) **Primary Key**: orderID **Foreign Key**: customerID, references **Customer** (customerID) **Foreign Key**: driverID, references **Driver** (driverID) **Foreign Key**: superMktID, references **Supermarket** (superMktID)
Driver (driverID, custProximity) **Primary Key**: driverID **Foreign Key**: None
Feedback (feedbackID, userExperience, groceryFreshness, deliveryService) **Primary Key**: feedbackID **Foreign Key**: orderID, references **Order** (orderID)
Payment (paymentID, receipt, orderDetail, deliveryAddress, totalCost, paymentConfirm) **Primary Key**: paymentID **Foreign Key**: orderID, references **Order** (orderID)
OrderTracking (orderTrackingID, orderStatus, pickupDetail, deliveryDetail) **Primary Key**: orderTrackingID **Foreign Key**: None

***Task 3**: Identify sensitive data and security considerations*

As part of the **Australian Privacy Principle** (APP), the company must take steps to protect the information from misuse, or prevent unauthorised access and modifications made to the individual's data to ensure that the company protects the data of both staff and customers.

In the Conceptual Relation Data Model (CRDM), the data that needs to be protected is within the **customer** entity, which is **email**, **mobileNo**, **socialProfile**, **fName** and **lName**, as well as the **deliveryAddress** which is considered a **personal** data that shows every information of the customer.

The next data to protect is the **financial** information and the data attributed to this is from the **payment** entity of the database that has the **paymentID**, **orderDetail**, **receipt**, and **totalCost** covered by the customer to guard the payment of the customers involved in the application of FreshCart.

The final data to maintain protection is the **health** of the customer as it can be used to acquire the information of a customer's **preference**, **dietPreference** from both the **order** entity and **customer** entity. Furthermore, many other data involved in health can be used to exploit the needs of the individual.

The **bullet points** below will explain which type of protection will be used for each data type (**personal**, **financial**, **health**):

1. **Personal Data**

- **Attributes:** *email, mobileNo, socialProfile, fName & lName*
  - **Protection**: *Anonymisation/Pseudonymisation*

Anonymisation allows the attributes to remain hidden preventing the customer's information from being displayed by masking the data

that the customer has given (e.g xxxx@xxx). This helps comply with '**Australian Privacy Principle – 2**' to remain anonymous and establish pseudonymity to allow the company to comply with the required regulations and uphold the individual's data anonymity.

2. Financial Data

- **Attributes**: paymentID, orderDetail, totalCost, receipt
- **Protection**: Encryption

Encryption prevents unauthorized access to financial details within an individual's data through encryption, which makes it only accessible by the primary owner of their paymentID as an example. The regulations from **APP** also apply for encryption as it provides confidentiality by hiding the data.

3. Health Data

- **Attributes**: preference, dietPreference
- **Protection**: Access Control

Access Control ensures that the individual will be the only one to have control or view over their information which in this case relates to the health preferences of the customer, which prevents sensitive data from being manipulated (diet preference or health preference). This makes it less likely for the data to be given to other individuals who can use this data to exploit the customer.

References:

Office of the Australian Information Commissioner n.d., *Australian Privacy Principles*, *Commonwealth of Australia*, viewed 30 July 2026, < https://www.oaic.gov.au/privacy/australian-privacy-principles>.

Privacy Act 1988 (Commonwealth of Australia), viewed 30 July 2026, < https://www.legislation.gov.au/C2004A03712/latest/text>.